FUTURISTIC & INTERACTION DESIGN

BY : Deepak, Mohit, Kunal, Nitish, Tanuj

End term exam

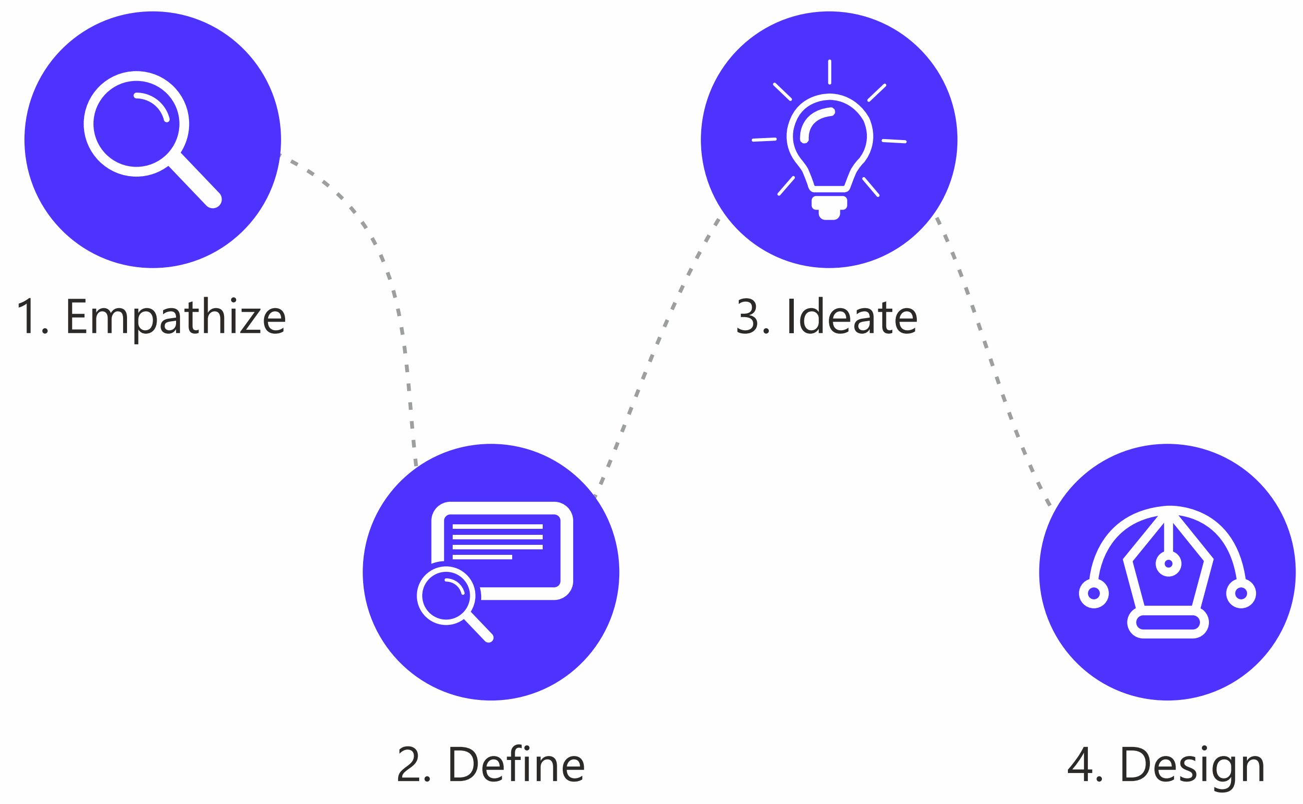
About

Our mission is to reduce the time and workload of traffic police officers at checkpoints. To achieve this, we have designed a AR headset tailored to support them in performing their duties more efficiently and effortlessly.

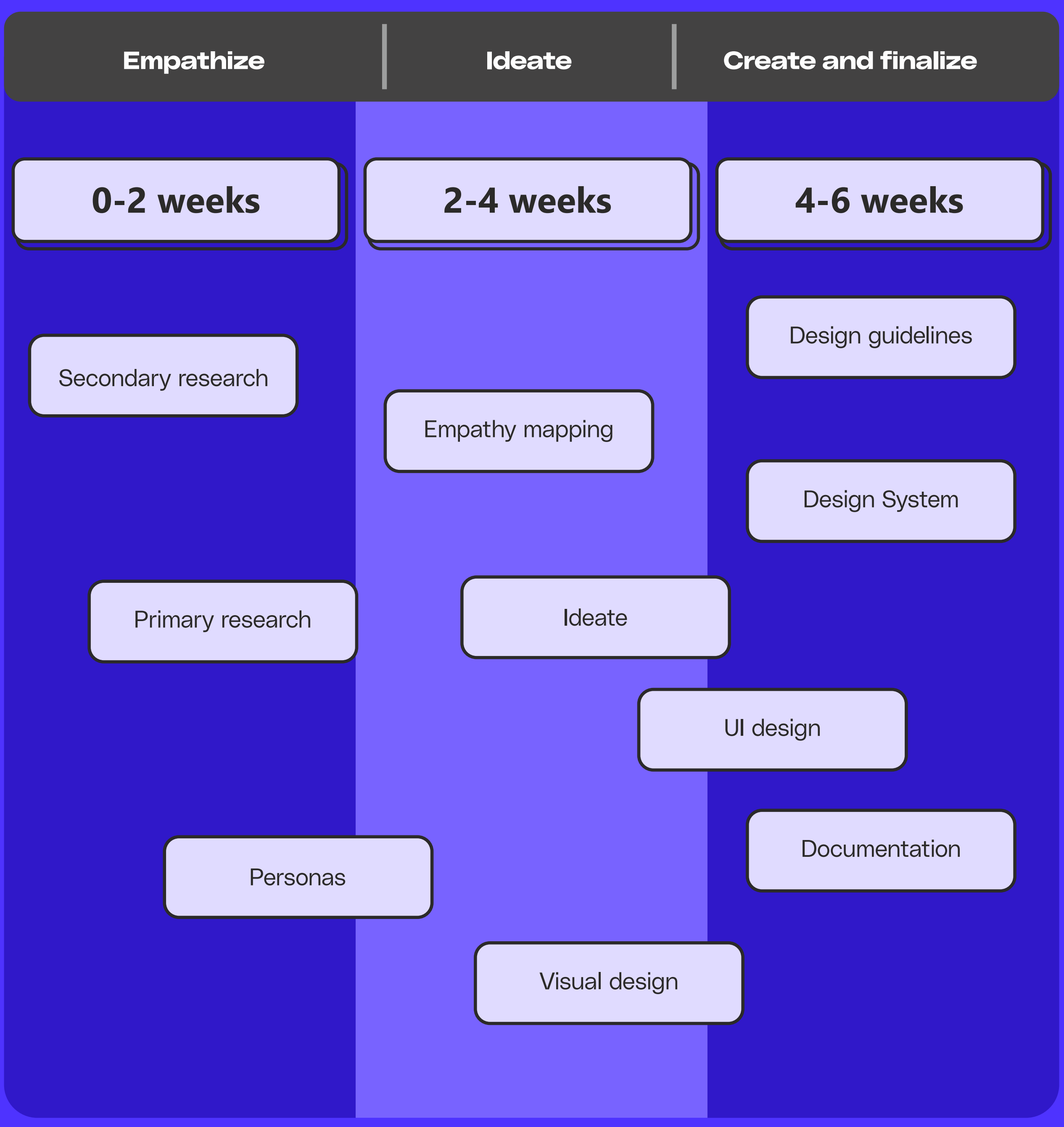
UI/UX design · Design thinking process · Ideation · Prototyping · User testing

Industry	Device	Timeline	Tools used
AR industry	Physical/digital	6 weeks	Ai Ps

How
did you do it ?



Project timeline



Empathize	Ideate	Create and finalize
0-2 weeks	2-4 weeks	4-6 weeks
Secondary research	Empathy mapping	Design guidelines
Primary research	Ideate	Design System
Personas	Visual design	UI design
		Documentation

Secondary research

Traffic police in densely populated cities like Delhi face numerous challenges while enforcing laws related to: (PRS India, 2021)

- Drunk driving checks
- Weapon seizures
- Seat belt
- Document verification

Challenges in Drunk Driving Enforcement

Public Resistance & Non-Cooperation: Many drivers refuse breathalyzer tests, leading to altercations. (Source: The Times of India, 2023)

Issues in Weapon Checks

Safety Risks: Traffic police are not always equipped to handle armed suspects, leading to dangerous encounters. (The Hindu, 2023)

Lack of Coordination with Local Police: Traffic cops often lack real-time criminal databases, making weapon detection difficult. (PRS India, 2021)

Seat Belt & Helmet Enforcement Problems

Evasive Tactics: Drivers lower seat belts when approaching checkpoints, making enforcement harder. (India Today, 2022)

High Non-Compliance Rates: Despite fines, many two-wheeler riders avoid helmets, and car passengers skip seat belts. (Road Transport Ministry Report, 2023)

30%

of suspected drunk drivers refuse testing, leading to legal complications. (The Indian Express, 2022)

05%

of random checks lead to weapon seizures due to lack of intelligence inputs. (NCRB Report, 2022)

THE HINDU

12 traffic police personnel were injured in altercations during weapon checks in 2022. (The Hindu, 2023)

65% of two-wheeler riders in Delhi do not wear helmets. (Ministry of Road Transport, 2023)

Despite **1.2 million** e-challans issued in 2023, repeat offenders remain high. (Delhi Traffic Police, 2024)

Delhi has only **6,000 traffic police** for over 11 million registered vehicles. (Delhi Transport Dept., 2023)

Only **10%** of Delhi traffic police have body cameras, leading to disputes. (India Justice Report, 2023)

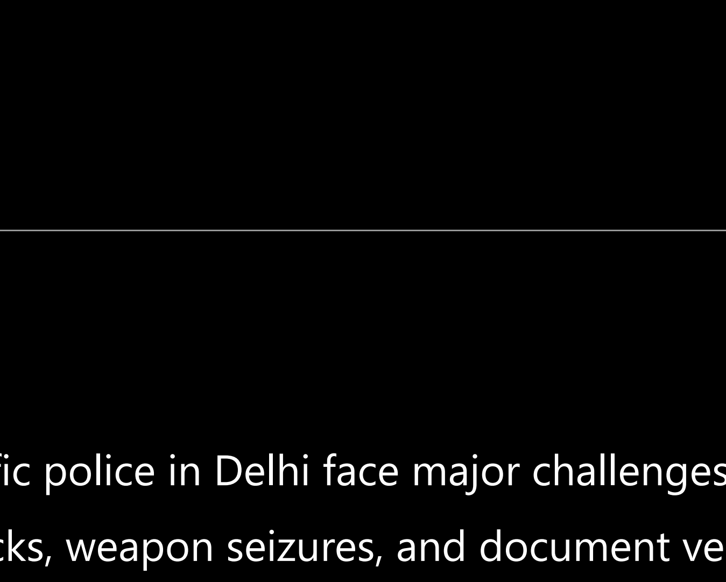
60% of fatal accidents at night go unchecked due to insufficient lighting. (Delhi Traffic Police, 2023)

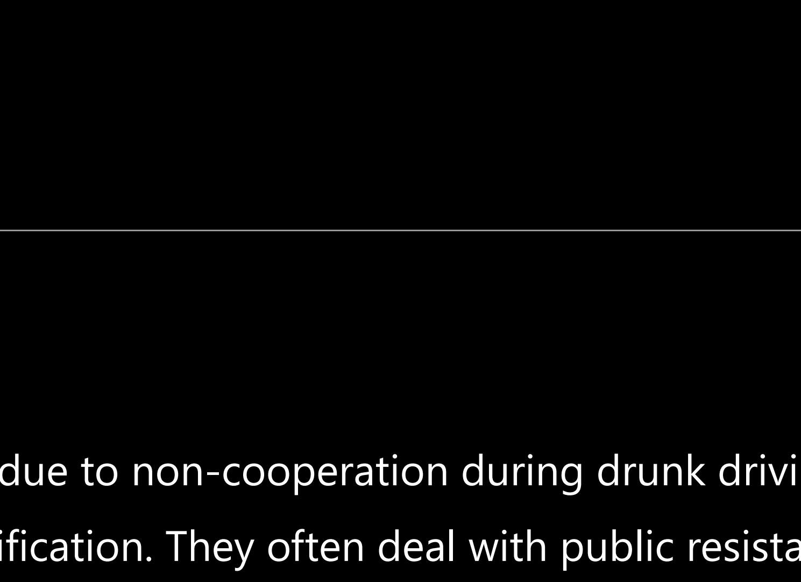
In 2024, the Delhi Traffic Police issued over **74 lakh traffic challans** and notices, with approximately **51.41 lakh (about 69%)** generated through automated traffic cameras. These camera-based notices primarily addressed violations such as red light jumping and speeding. (The week)

Under Delhi's Excise Rules, a private individual may legally carry up to 9litres of spirits (whisky, vodka, rum, etc.) and up to 18 litres of beer, wine or cider in a personal vehicle. (Source : Delhi Excise Department)

THE TIMES OF INDIA

Regarding illegal arms and liquor seizures, during a 16-day period from January 7 to 22, 2025, ahead of the Delhi assembly elections, the Delhi Police registered 504 cases for model code violations. In this crackdown, they seized 270 illegal arms, 372 cartridges, liquor worth ₹1.3 crores, and arrested 17,879 individuals for preventive actions .

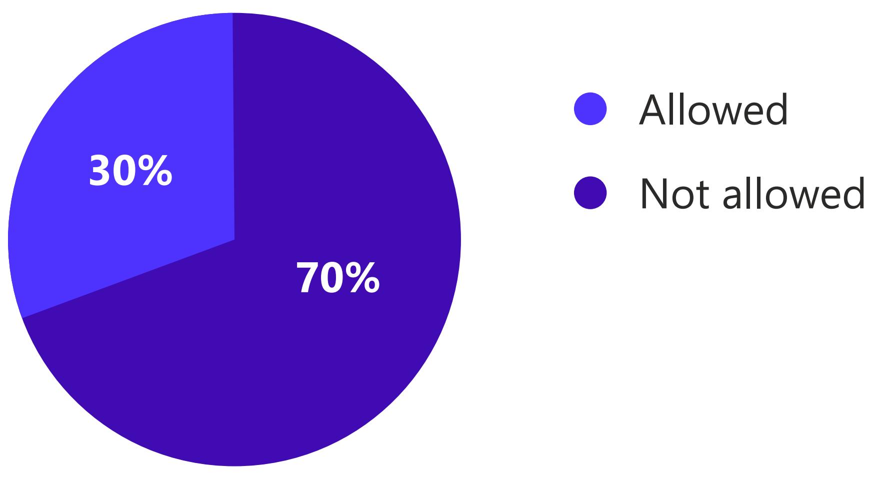




Traffic police in Delhi face major challenges due to non-cooperation during drunk driving checks, weapon seizures, and document verification. They often deal with public resistance, legal delays, safety risks, and outdated equipment. Staff shortages and lack of tech support like body cams and functioning breathalyzers worsen enforcement.

Primary research

We approached about 13 traffic officers for interviews, but only four could spare the time; the rest were overwhelmed, constantly stopping rule-breakers and couldn't step away from their duties.



Q: Describe a typical day at a checkpoint.
We set up the barricades by 8 AM. Then we monitor traffic, stop random vehicles, check documents, and ensure helmets and seatbelts are in use. During peak hours, it becomes chaotic. (Head constable, age 42 and 15 year experience)

Q: What's most draining about the job?
The public's attitude. People think we're here just to fine them, not to ensure safety. That lack of respect is tiring. (constable, age 29 and 5 year experience)

"Almost every weekday morning. Within ten minutes of setting up the barricades the queue stretches two hundred metres. Drivers lay on their horns, two-wheelers squeeze between cars, and I'm juggling paperwork while waving people forward. It's a constant battle to keep tempers—and traffic—under control."(Head constable, age 42 and 15 year experience)

"Yes—twice in the last five years. One was a hatchback with 40 bottles of illicit whisky hidden under cartons of fruit; the driver looked nervous, so I asked him to open the boot. The second time we found a small country-made pistol in a glove box during a late-night DUI check."(Head constable, age 42 and 15 year experience)

Q: Common problem at checkpoints? (Driver)
We are stopped just because we drive heavy vehicles. It delays us unnecessarily.

Do the red-light/traffic-violation cameras make your work at checkpoints safer or more efficient? How?
"Yes, Because fixed cameras catch many red-light jumpers and speeders automatically, we can position fewer men in the middle of busy junctions"

do cameras can check liquor or weapon smuggling?
"As useful as red-light cameras are, they only snap the number plate and catch signal or speed violations; they can't look inside a car. If someone is carrying liquor cartons or a hidden weapon, the camera won't see it—we still have to pull the vehicle over and inspect it ourselves."

Both drivers and officers see checkpoints as stressful bottlenecks, yet for opposite reasons:
drivers fear delays and confrontation, while officers are overwhelmed by real-time violations and personal safety risks. Cameras ease routine traffic offences, but they neither reveal hidden contraband nor reduce the face-to-face workload that causes queues and conflict—highlighting a gap for tools that can screen vehicles quickly and securely before human interaction begins.

Define

Traffic-police officers at checkpoints must manually stop and inspect vehicles—especially commercial ones—for documents, safety gear, alcohol, or contraband. Each hand-run check takes several minutes, quickly backing up traffic into long queues. The resulting congestion forces officers to rush inspections, heightens driver frustration, and exposes the officers to safety risks, ultimately preventing them from enforcing laws thoroughly or efficiently.

Scenario

Rajiv Sharma, 42 (Primary user)

It’s 8:00 a.m. at the Delhi-Gurugram border. RJ, a traffic officer, places cones on the road and starts checking cars. In just ten minutes, a long line of honking vehicles forms behind him, stretching 200 metres.He stops a small truck, asks the driver for his license and pollution papers, and opens the back of the truck. As RJ checks the crates for hidden liquor, the waiting drivers grow impatient. Some of them start moving forward, blocking the lane meant for other vehicles.RJ calls for help on his radio, but the other officers are stuck in the same traffic jam. With loud horns blaring around him and time running out, RJ quickly finishes checking the truck and lets it go. When he turns around, he sees that a motorcyclist has passed by without wearing a helmet.The process becomes slower and more frustrating for everyone.

Aman Verma, 31 (Secondary user)

Aman leaves home at 8:10 a.m. for his 9:30 meeting in Gurugram. Google Maps shows smooth traffic at first. But at 8:45, he reaches a checkpoint where everything slows down. The line moves slowly, then stops completely.He sees an officer checking a delivery van and opening every box. Ten minutes pass. His smartwatch reminds him: “Meeting starts in 30 minutes.”Sitting in his car with the AC on and fuel going down, Aman switches between checking emails and pressing the clutch.Finally, he’s called forward. He shows his digital licence and insurance. The officer takes a quick look and waves him on.As Aman drives back into traffic, he starts thinking about what to say to his manager—he’ll have to join the meeting from his car.

Persona

Rajiv Sharma, 42 (Primary user)

Inspector Rajiv “RJ” Sharma is a 42-year-old Delhi traffic officer who has spent 18 years taming the city’s busiest checkpoints. Driven to keep roads safe, he battles queue chaos, safety risks, and equipment gaps while leading a six-person enforcement squad.

Pain points

Long ques of vehicles at checkpoints.

Not able to check vehicles properly because of long ques.

It’s get difficult to check every vehicles due to heavy traffic.

misbehave and abuse

Goals

Minimize congestion at check points.

Do checking with efficiency and effectively.

Safety of his and his team

Needs

Body-cam for personal safety, evidence, and transparency.

Equipments which reduce the

Aman Verma, 31 (Secondary user)

AmanVerma, 31, is a senior business analyst who drives from SouthDelhi to his Gurugram MNC every weekday. Punctual and tech-savvy, he endures a 10-to-15-minute checkpoint queue at the Delhi-Gurugram border that chips away at his time, fuel—and patience—before each workday begins.

Pain points

Lost Time: 10–15min daily queue

Wait length varies; can’t reliably plan arrival.

Goals

Arrive at office on time & calm for morning meetings.

Reduce monthly fuel spend and wear -and-tear on car.

Get quick, fair checks

Needs

Transparent rules on who gets stopped and why.

Predictable checkpoint flow—or at least real-time wait updates.

Empathy map

Say

“Queues build in ten minutes—keep moving!”

“Cameras help, but they can’t spot hidden items.”

“Please cooperate—helmet, seat-belt, that’s all.”

Think

“If I had faster tools, I could clear cars quicker.”

“Public trust depends on fair, transparent policing.”

Does

Waves traffic, checks docs, radios plate numbers, settles disputes.

Waves traffic, **stops and inspects vehicles—especially commercial vans and trucks

Feel

Frustrated by equipment gaps; hopeful about new tech.

Alert, pressured, protective of team.

Say

“Why is this lane crawling again?”

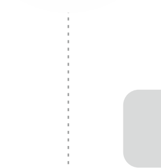
“I follow the rules—why can’t we breeze through?”

“Tech should make this smoother.”

Think

“If I’m late, my boss will notice.”

Why here is so much traffic at same point.



Does

Crawling the car

showing documents to the officers.

Feel

Anxious, impatient, stressed.

Hopeful that smarter systems will ease the queue.

How might we statements

How might we speed up checks so lines at checkpoints stay short?

How might we keep officers safe from getting hurt on the road?

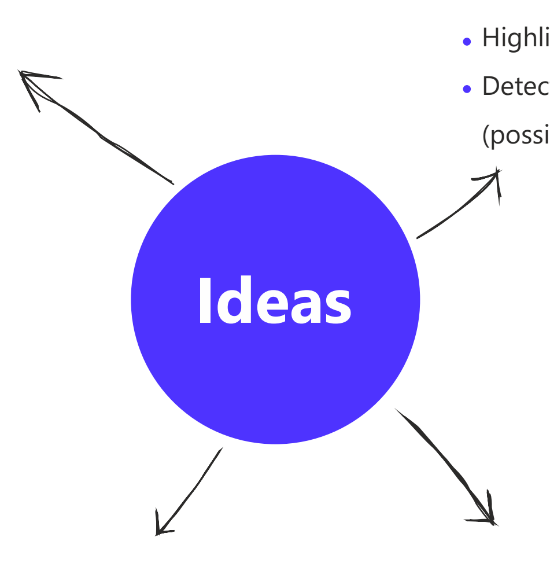
How might we spot the truly suspicious cars first, so only they get stopped?

How might we help drivers stay calm and know what’s happening while they wait?

How might we give police new tools and instant info to do their job faster?

How might we help officers work faster and get better results at the checkpoint?

Ideation



Predictive Threat Detection (AI Integration)

AI overlays in AR that suggest "risk score" based on:

- Vehicle history
- Time of day
- Location-based crime data

AR-Driven Smart Scanning

Use AR headset with integrated license plate scanner, face recognition, and infrared scanner to:

- Instantly fetch vehicle records
- Identify past offenses
- Highlight expired documents visually on the HUD
- Detect hidden compartments or abnormal heat signatures (possible contraband)

Real-Time Offense Database Sync

Seamless integration with national RTO, police, and criminal databases for:

- Live updates
- Pending fines
- Vehicle theft alerts
- Arms or liquor transport alerts

Officer Safety Enhancements

- SOS gesture or voice trigger in AR headset
- Auto-recording of interactions for legal and training purposes
- Red-flashing visual alerts

Final idea

Our final concept is an advanced AR headset designed specifically for traffic police officers, equipped with features such as:

- Real-time speed detection of moving vehicles
- Automatic challan alerts by scanning and identifying vehicle number plates
- Officer safety alerts in case of verbal abuse, physical threats, or emergencies
- X-ray-style vehicle scanning using advanced sensors to detect hidden items
- Instant alerts for stolen vehicles based on number plate recognition
- Detection of illegal transport such as weapons or liquor, triggering real-time flags to officers

Ideation to Reality

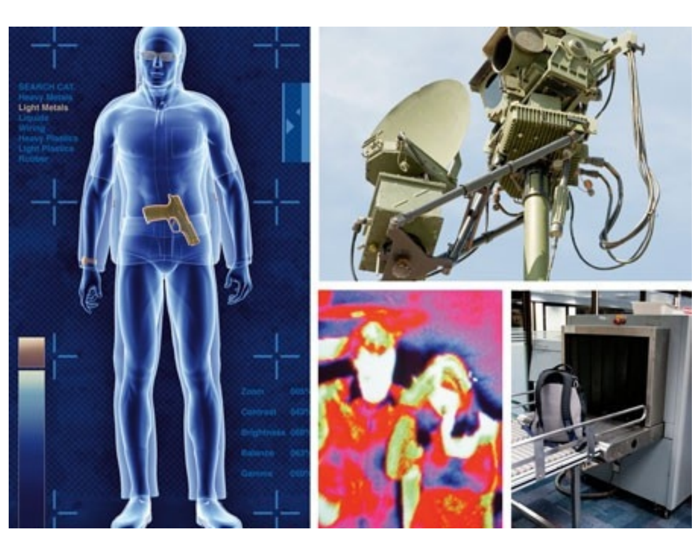
Speed Cameras Detection Technology

Speed cameras utilize advanced detection technologies to identify and capture speeding vehicles.

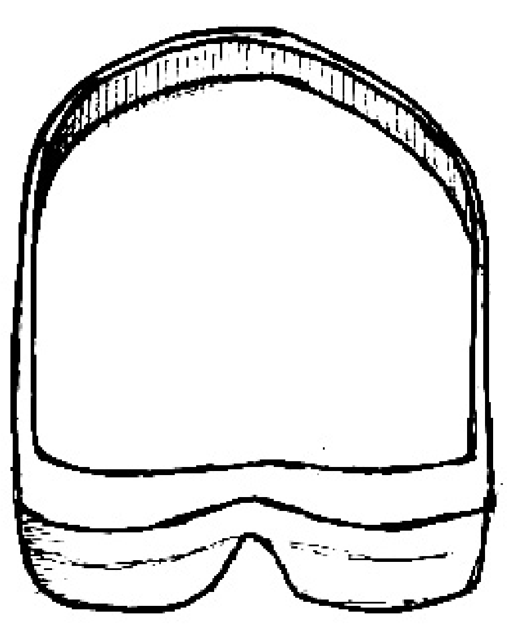
- **Radar Technology**: Most speed cameras employ radar (Radio Detection and Ranging) technology. They emit radio waves and measure the time it takes for the waves to bounce back after hitting a vehicle. By calculating the time taken, the camera determines the distance and speed of the vehicle.

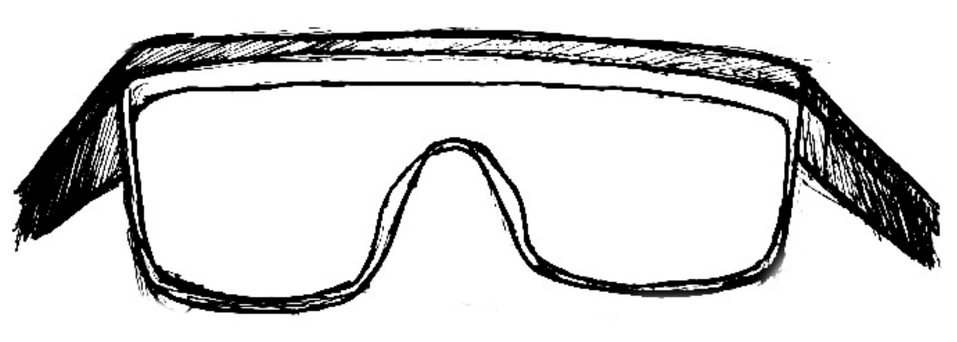
- **Laser Technology**: Some speed cameras utilize laser technology, emitting pulses of laser light and measuring the time it takes for the light to reflect off the vehicle. Similar to radar, the camera calculates the distance and speed based on the time measurement. (Source : Linked in)

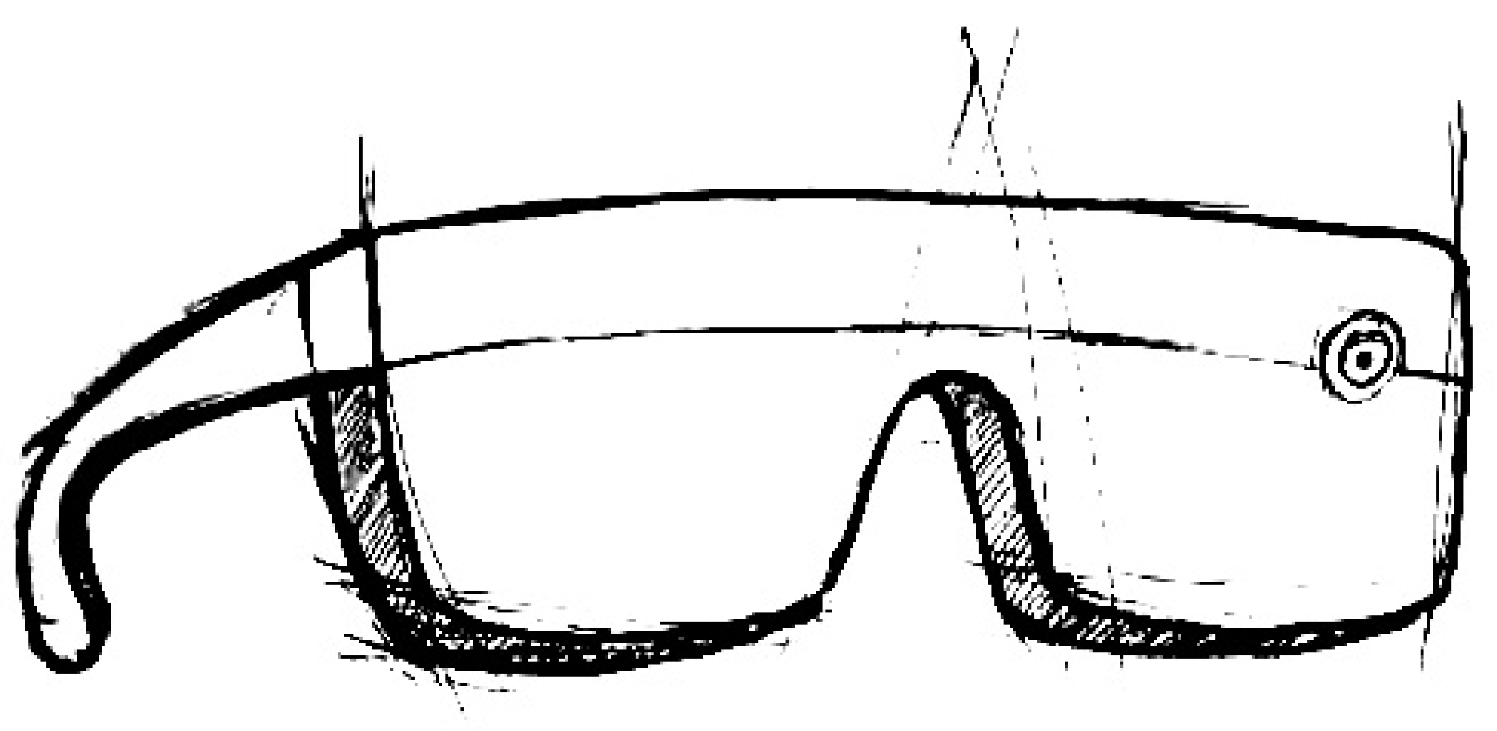
LIDAR-lite V3

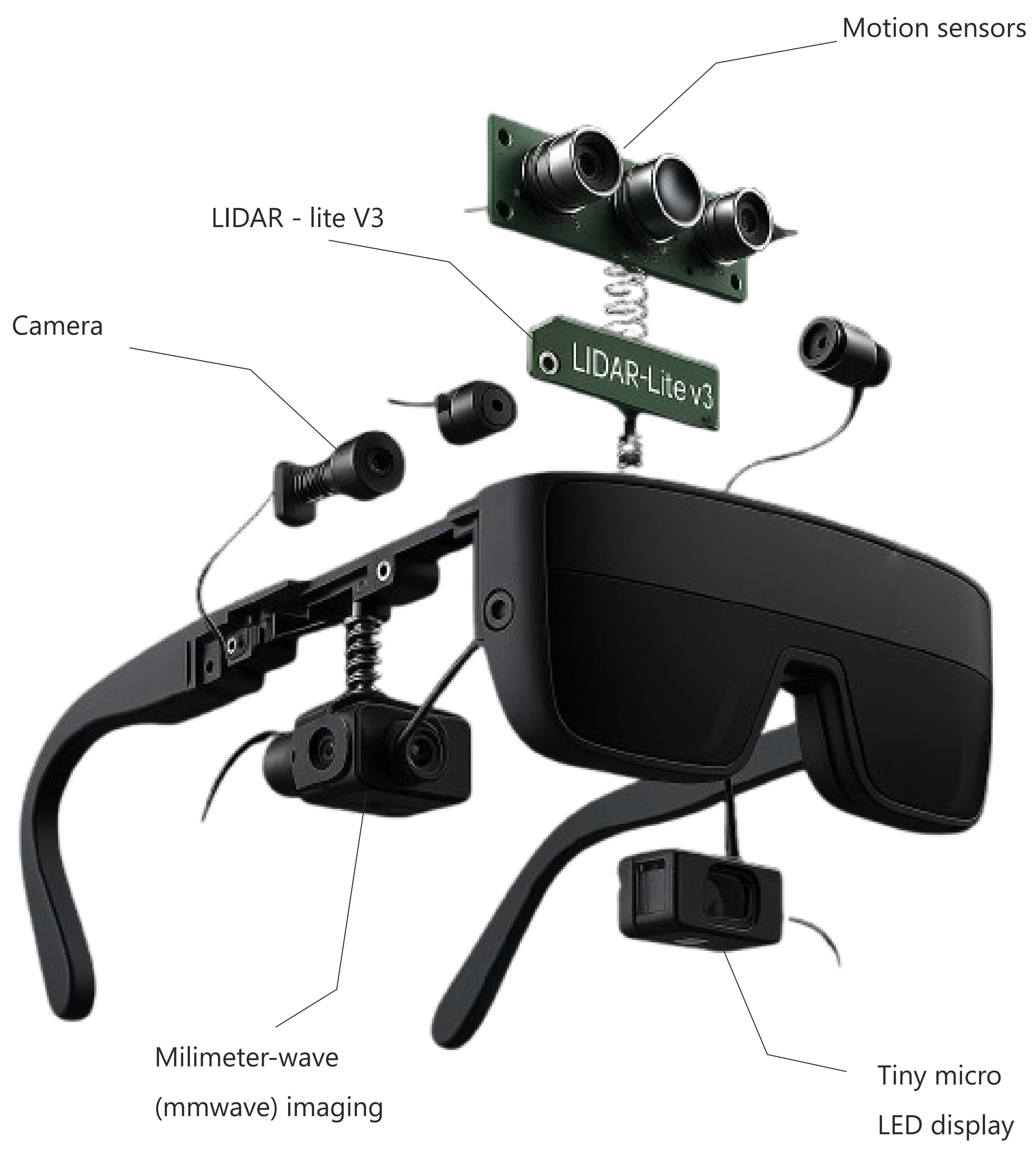


Millimeter-Wave (mmWave) Imaging









VISIONCOP

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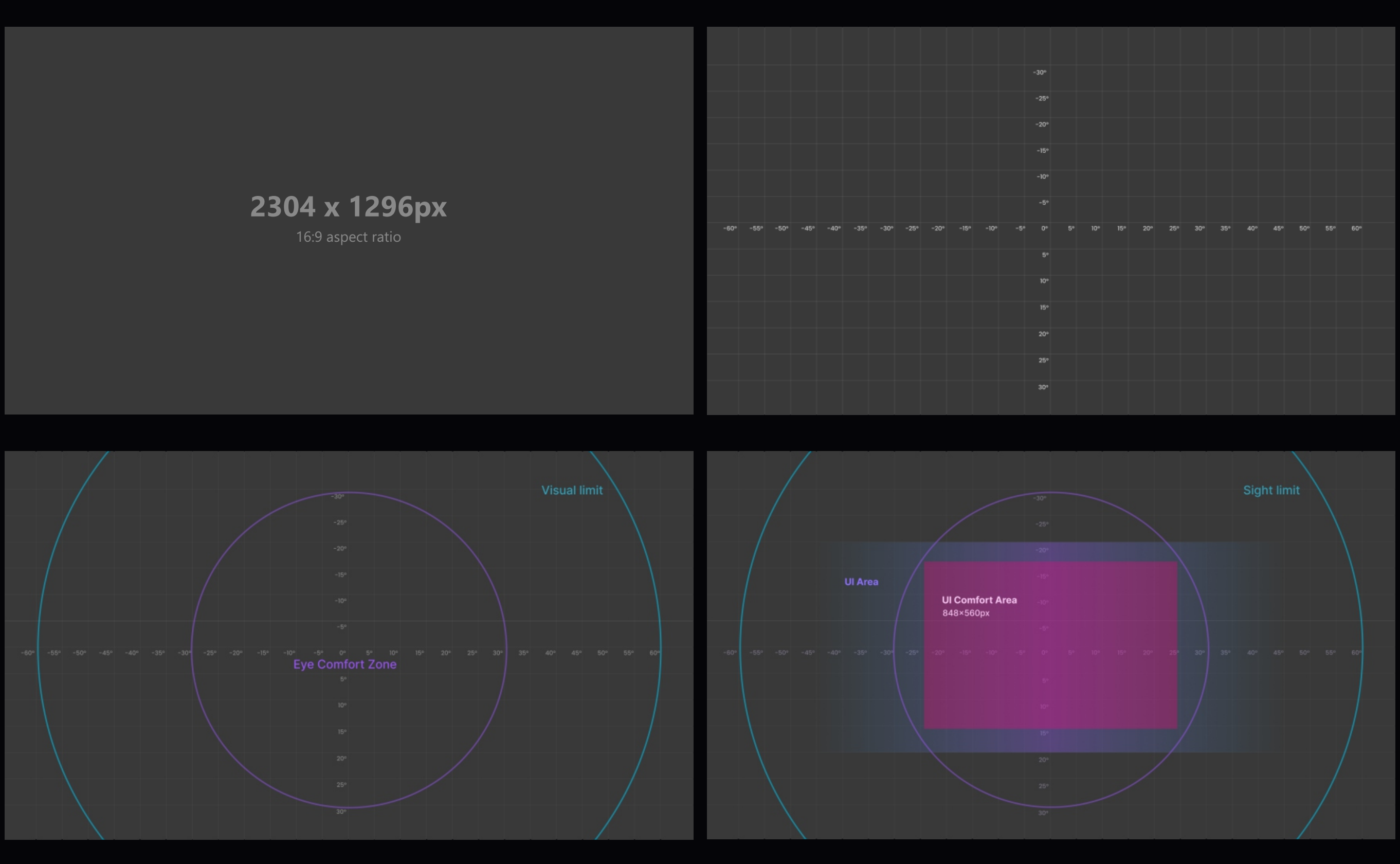
Design
guidelines



Design
System

Typography

Inter

Aa

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

a b c d e f g h i j k l m n o p q r s t u v w x y z

1 2 3 4 5 6 7 8 9 0

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

a b c d e f g h i j k l m n o p q r s t u v w x y z

1 2 3 4 5 6 7 8 9 0

Extra large (Bold, 44)

Header 1 (Bold, 32)

Header 2 (Bold, 26)

Header 3 (Bold, 20)

Header 4 (Bold, 16)

Title (Medium, 16)

Body (Regular, 16)

Body 2 (Regular, 14)

Body 3 (Regular, 12)

Button large (Semi bold, 18)

Button medium (Semi bold, 16)

Button small (Semi bold, 14)

Colors

Primary colors

Light #B48CF5	Main #9754F0	Dark #663CAA

Success

Light #60A774	Main #127C2F	Dark #0D5821

Error

Light #E07B7B	Main #D13A3A	Dark #942929

Warning

Light #FFBF72	Main #EB9D3F	Dark #B57120

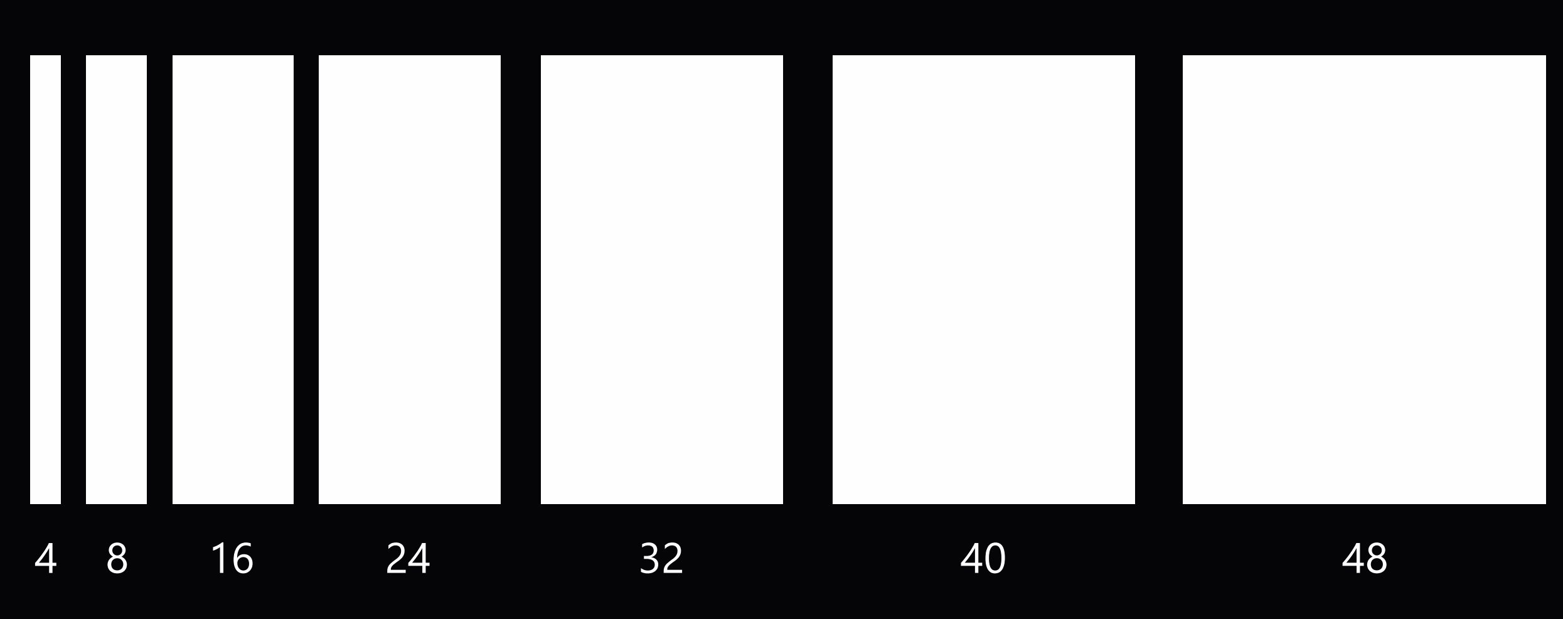
Info

Light #63B4FF	Main #168FFF	Dark #1066B5

Surfaces

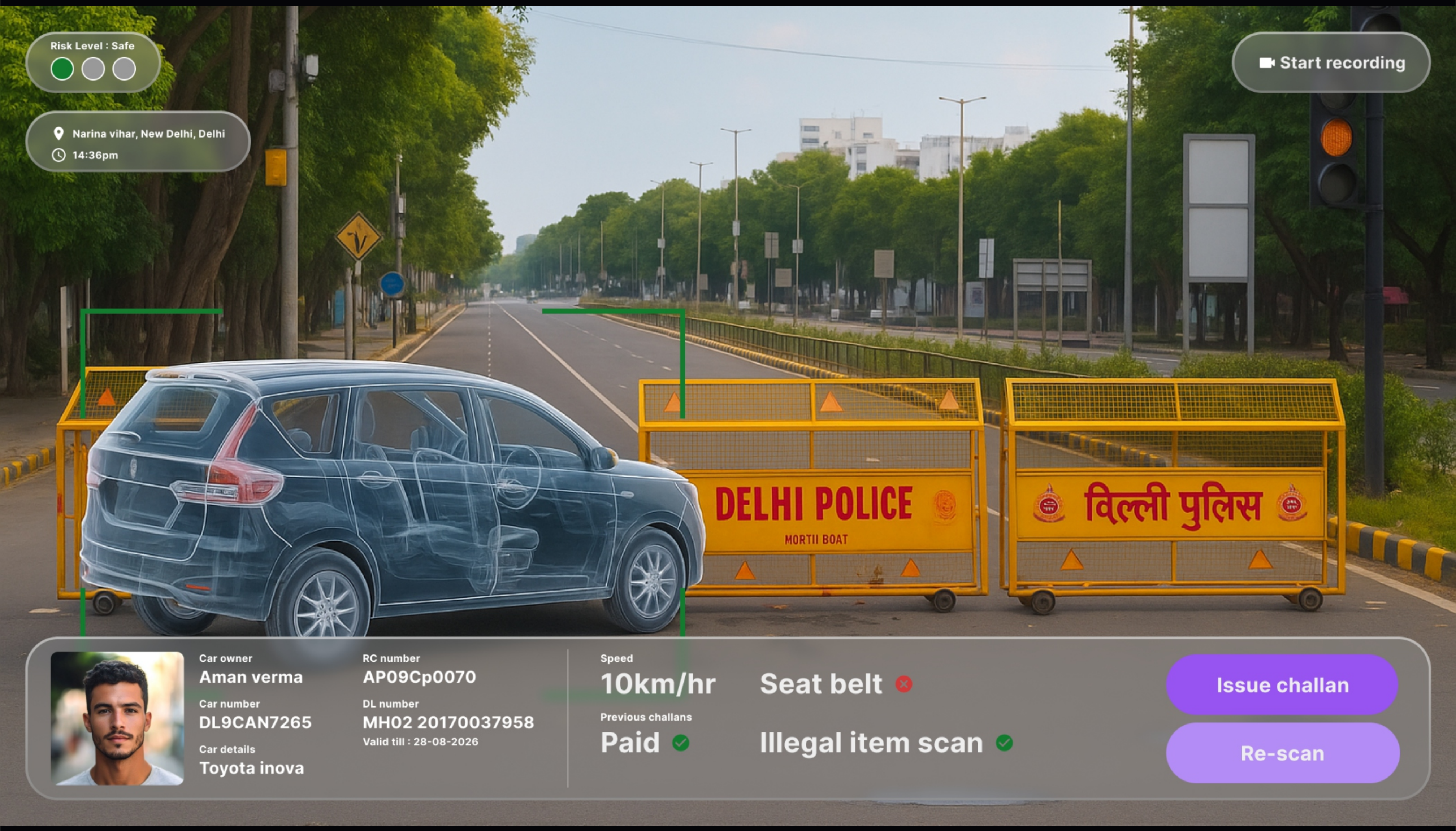
Surface 1 #8C8C8C Opacity - 50%	Surface 2 #4F4F4F Opacity - 40%	Surface 3 #303030 Opacity - 30%

Spacing



Icon library

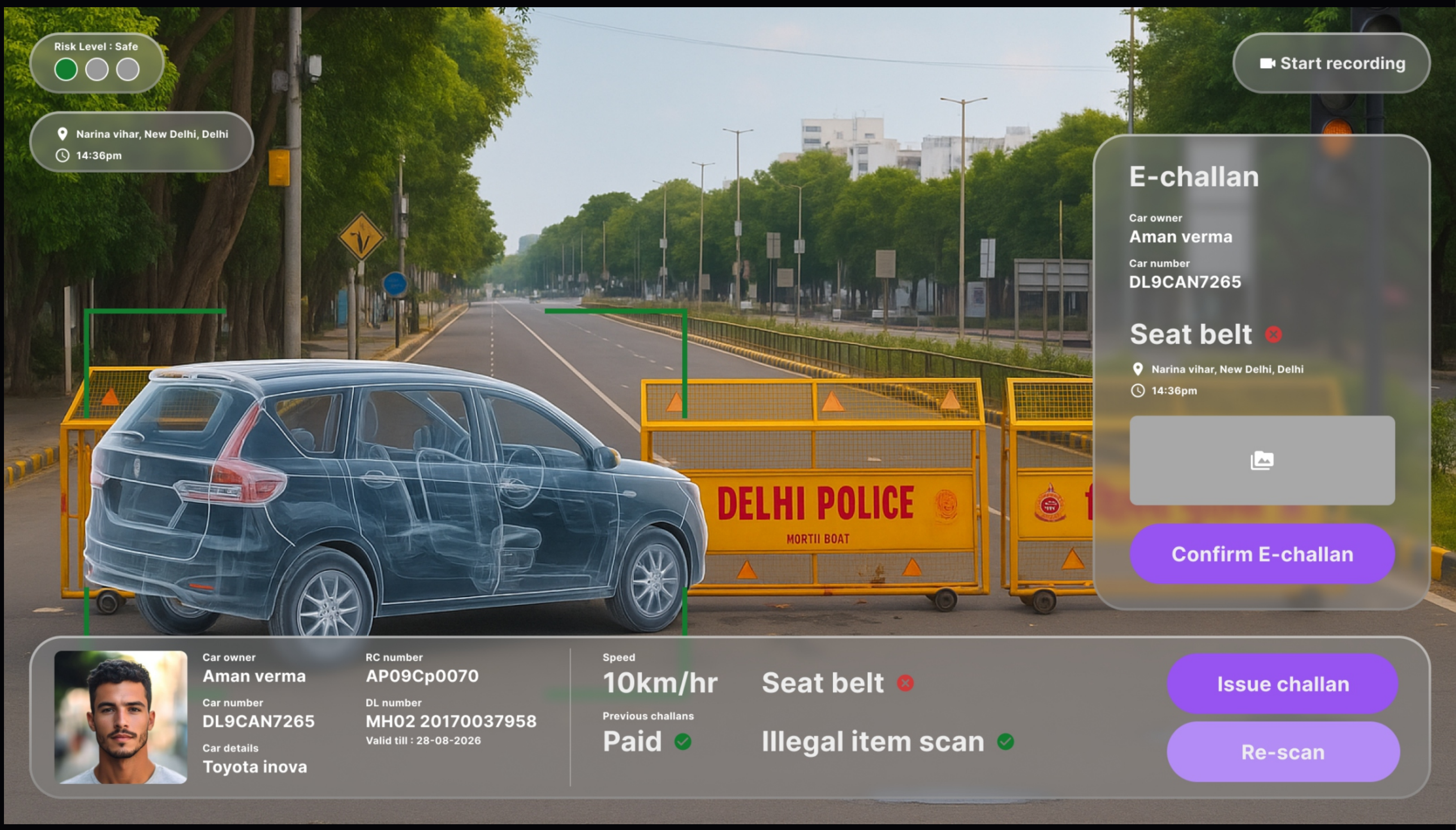
UI design



Risk Level : Safe ● ○ ○

Narina vihar, New Delhi, Delhi
14:36pm

Car owner: Aman verma
Car number: DL9CAN7265
Car details: Toyota inova
RC number: AP09Cp0070
DL number: MH02 20170037958
Valid till : 28-08-2026
Speed: 10km/hr
Previous challans: Paid ✓
Seat belt ✗
Illegal item scan ✓



Risk Level : Safe ● ○ ○

Narina vihar, New Delhi, Delhi
14:36pm

E-challan
Car owner: Aman verma
Car number: DL9CAN7265
Seat belt ✗
Narina vihar, New Delhi, Delhi
14:36pm

Car owner: Aman verma
Car number: DL9CAN7265
Car details: Toyota inova
RC number: AP09Cp0070
DL number: MH02 20170037958
Valid till : 28-08-2026
Speed: 10km/hr
Previous challans: Paid ✓
Seat belt ✗
Illegal item scan ✓

Thank you